

Classifying Kidney Transplant Rejection from Biopsy RNA

Transplant rejection and kidney biopsy

- **Rejection:** The recipient's immune system attacks the donated kidney.
- **Kidney biopsy:** A small tissue sample collected from the kidney for examination.
 - **Histology:** A microscopic examination that reveals injury and inflammation, which can support a rejection diagnosis.
 - **Molecular measurements:** Counts of selected RNA types that reflect gene activity in the mix of cells in the tissue.

Purpose of this project

Whether rejection is present can affect decisions about further treatment to suppress the immune system.

Model comparison

How do logistic regression and CatBoost compare in classifying any recorded rejection versus no rejection from B-HOT biopsy RNA, particularly in missed cases and incorrect flags?

Software engineering

Make preprocessing, training and evaluation reproducible. Build a tested scoring service and prototype application that let a user submit a specimen's RNA counts and inspect its model score, threshold and rejection flag.

Clinical and research context

Banff diagnostic framework

Validated biopsy transcript tests have a defined role in antibody-mediated rejection assessment.

Published B-HOT study Zhang, 2024

Four-class study compared regression and boosting. Selected LASSO for similar accuracy with fewer features.

Why both kinds of error matter

Biopsy findings help guide treatment for rejection. Missed rejection can leave kidney injury untreated. Unnecessary treatment can worsen an infection.

What the dataset contains

1,395 biopsy specimens, with one row per specimen in each table.

RNA counts

Column	Example
Specimen ID (key)	GSM6510425
IFNG	4
GUSB (reference)	260
Other RNA counts	...

1 : 1

Same ID

Specimen information

Column	Example
Specimen ID (key)	GSM6510425
Recorded diagnosis	No rejection
Study group	Discovery
Other details	...

- Includes 1,193 transplant biopsies and 202 native-kidney controls.
- The study does not document separation by patient or referring center between study groups.

Outcome: rejection versus no rejection

The model combines the three rejection categories in the public dataset.

Recorded diagnosis	Meaning	Model label
Antibody-mediated rejection	Antibodies: immune proteins	Rejection
T-cell-mediated rejection	T cells: immune cells	Rejection
Mixed rejection	Both rejection processes	Rejection
No rejection	No recorded rejection	No rejection

The authors excluded some diagnostic categories from the dataset.

The measurements used to build the features

770 RNA counts per specimen, one for each RNA type measured.

758

Model measurements

- Selected human and viral RNA measurements.
- Signals reflect gene activity and the mixture of cells in the biopsy.
- Normalized values become the model's inputs.

12

Housekeeping references

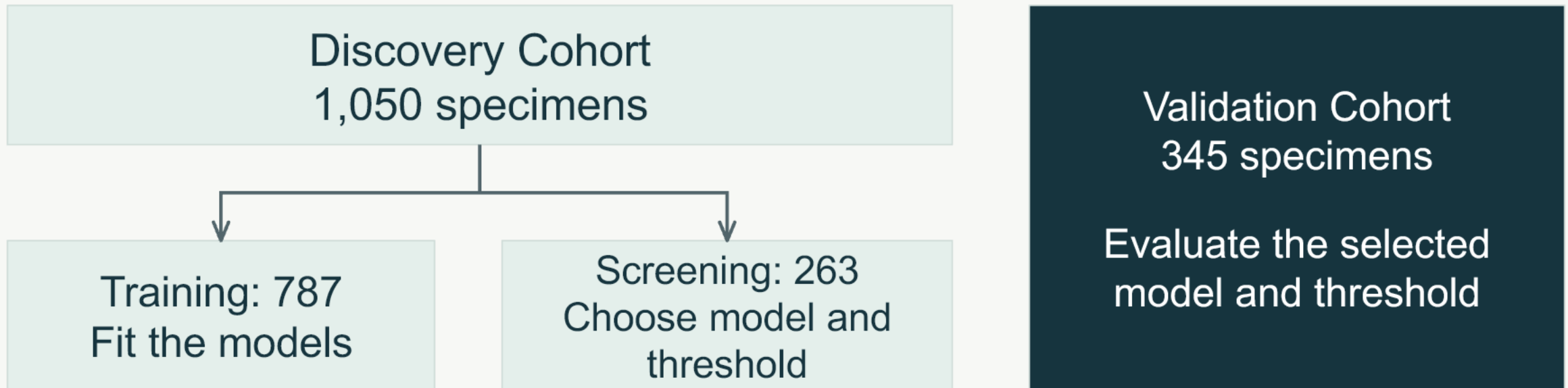
- Relatively stable RNAs provide a reference for each specimen.
- Help adjust for differences in overall measurable RNA input.
- Used for normalization, then excluded from the model.

Preparing counts for the model

For each of the 758 RNA counts	IFNG example
1. Start with the raw count	4
2. Calculate $\log_2(\text{count} + 1)$	2.322
3. Average $\log_2(\text{count} + 1)$ for the 12 reference RNAs	7.900
4. Subtract that reference	$2.322 - 7.900 = -5.578$

Development and evaluation specimens

- I used the authors' two study groups: discovery and validation cohorts.
- My training / screening split kept diagnosis proportions similar.



Models compared

Each model produces a score for rejection versus no rejection.

Model	Approach and rationale
Constant baseline	One score: the training rejection fraction
IFNG only (Logistic regression)	One immune-related RNA measurement as a simple benchmark
All RNA (Logistic Regression)	A weighted combination of 758 measurements. Regularization limits the size of the weights.
CatBoost	Can capture nonlinear RNA patterns and interactions. Shallow trees help limit overfitting.

Selecting the model and threshold

Recall

Correctly flagged rejection cases

All cases diagnosed as rejection

(False negatives are missed cases.)

Precision

Correctly flagged rejection cases

All rejection flags

(False positives are incorrect flags.)

Screening rule: detect at least 90% of recorded rejection cases, then minimize false positives. ROC-AUC breaks ties.

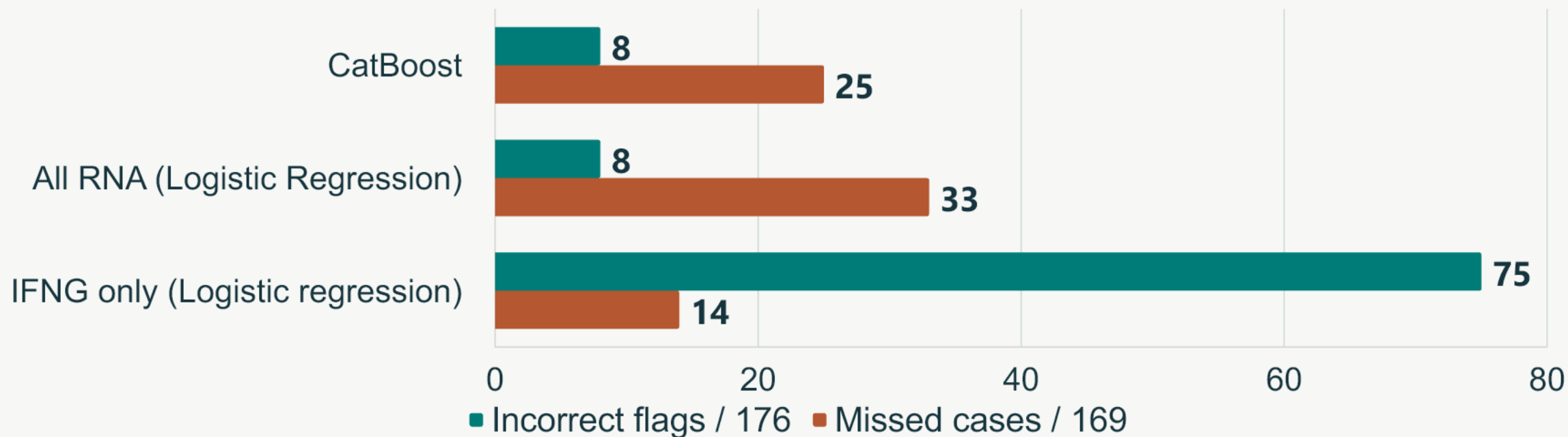
Screening result	CatBoost	All-RNA logistic
Rejection detected	157 / 174	157 / 174
False positives	5 / 89	6 / 89

One fewer false positive selected CatBoost.

The model and threshold then stayed fixed for evaluation.

Validation results

Same 345 specimens: 169 rejection and 176 no rejection



- Constant baseline: 0 missed cases, 176 incorrect flags. It flags everyone.

Across 20 discovery splits: CatBoost selected 13 times, logistic 7.

Software engineering

Reports, application and scoring service

- Inspect the evidence
- Score a public specimen
- Trace and check the calculation

What this project accomplished

Model comparison

CatBoost missed 8 fewer rejection cases than logistic, with 8 incorrect flags each. A dependable advantage remains uncertain.

Software engineering

Reproducible analysis and a tested scoring service, with a prototype application and checks against saved results.

Next: cross-validation within discovery.

Then investigate whether the score adds useful diagnostic information.